

CHAPTER 4

IDENTIFICATION OF FUNCTIONAL DEPENDENCIES, REDUNDANCY ANALYSIS, AND DATA ANOMALIES

4.1 INTRODUCTION

This chapter explains the **functional dependencies**, **redundancy analysis**, and **data anomalies** for the Etsy Marketplace Database System. Functional dependencies describe the relationship between attributes within a table and help identify how data is uniquely determined. Understanding these dependencies is essential for designing a well-structured database.

This chapter also examines redundancy and the possible data anomalies that may occur if the database is not properly normalized. Identifying these issues helps improve data consistency, reduce duplicate information, and maintain the integrity of the database.

4.2 CUSTOMER

Attribute	Description
Customer_ID	Unique customer identifier
First_Name	Customer's first name
Last_Name	Customer's last name
Email	Customer email
Password	Login password
Phone_Number	Contact number
Address	Delivery address

4.2.1 Functional Dependency

Customer_ID → **First_Name, Last_Name, Email, Password, Phone_Number, Address, City, State, Country, Postal_Code, Registration_Date**

Explanation:

- Each **Customer_ID** uniquely identifies one customer.
- All customer details depend only on **Customer_ID**.

4.2.2 Redundancy Analysis

- Customer information should be stored only once.

- Storing the same customer details in multiple orders would create duplicate data.
- Using **Customer_ID** as a foreign key in the Order table eliminates redundancy.

4.2.3 Data Anomalies

Insertion Anomaly

- Customer details cannot be stored if the system requires an order before creating a customer record.

Update Anomaly

- Updating a customer's phone number in multiple places may result in inconsistent data.

Deletion Anomaly

- Deleting the last order could accidentally remove customer information if stored together.

4.3 SELLER

Attribute	Description
Seller_ID	Unique seller identifier
Seller_Name	Seller's name
Shop_Name	Shop name
Email	Seller email
Password	Login password
Phone_Number	Contact number
Address	Business address
Rating	Average seller rating
Join_Date	Registration date

4.3.1 Functional Dependency

Seller_ID → **Seller_Name, Shop_Name, Email, Password, Phone_Number, Address, Rating, Join_Date**

4.3.2 Redundancy Analysis

- Seller details should be maintained in a single table.
- Products should reference the seller using **Seller_ID** instead of repeating seller information.

4.3.3 Data Anomalies

Insertion Anomaly

- Seller details cannot be added if product information is mandatory.

Update Anomaly

- Updating seller information in every product record may create inconsistencies.

Deletion Anomaly

- Deleting the last product may remove seller information.

4.4 SHOP

Attribute	Description
Shop_ID	Unique shop identifier
Seller_ID	Shop owner
Shop_Name	Name of the shop
Description	Shop description
Logo	Shop logo
Location	Shop location

4.4.1 Functional Dependency

Shop_ID → Seller_ID, Shop_Name, Description, Logo, Location

4.4.2 Redundancy Analysis

- Shop information should exist only once.
- Multiple products should reference the shop indirectly through the seller.

4.4.3 Data Anomalies

Insertion Anomaly

- A shop cannot be recorded if the seller does not exist.

Update Anomaly

- Updating the shop description in multiple tables creates inconsistency.

Deletion Anomaly

- Removing the seller may also remove shop information.

4.5 PRODUCT

Attribute	Description
Product_ID	Unique product ID
Seller_ID	Seller who listed the product
Category_ID	Product category
Product_Name	Product name
Description	Product description
Price	Selling price
Quantity	Available stock
Material	Product material
Color	Product color
Size	Product size
Image_URL	Product image
Listing_Date	Product listing date

4.5.1 Functional Dependency

Product_ID → **Seller_ID**, **Category_ID**, **Product_Name**, **Description**, **Price**, **Quantity**, **Material**, **Color**, **Size**, **Image_URL**, **Listing_Date**

4.5.2 Redundancy Analysis

- Product information should be stored once.
- Order tables should reference **Product_ID** instead of duplicating product details.

4.5.3 Data Anomalies

Insertion Anomaly

- Product cannot be inserted without a valid seller and category.

Update Anomaly

- Updating product price in multiple tables causes inconsistency.

Deletion Anomaly

- Removing product records may affect historical order information if not properly designed.

4.6 CATEGORY

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Description	Category description

4.6.1 Functional Dependency

Category_ID → **Category_Name, Description**

4.6.2 Redundancy Analysis

- Category names should not be repeated for every product.
- Products should reference **Category_ID**.

4.6.3 Data Anomalies

Insertion Anomaly

- Category cannot be added if products are required first.

Update Anomaly

- Renaming a category in multiple product records creates inconsistency.

Deletion Anomaly

- Deleting the last product could remove category information.

4.7 ORDER

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer placing the order
Order_Date	Date of purchase
Total_Amount	Total bill amount
Order_Status	Pending, Shipped, Delivered
Shipping_Address	Delivery address

4.7.1 Functional Dependency

Order_ID → **Customer_ID, Order_Date, Total_Amount, Order_Status, Shipping_Address**

4.7.2 Redundancy Analysis

- Order details should be stored once.
- Order items should reference **Order_ID**.

4.7.3 Data Anomalies

Insertion Anomaly

- Order cannot exist without a customer.

Update Anomaly

- Updating order status in multiple tables causes inconsistencies.

Deletion Anomaly

- Removing an order may remove associated payment and shipping details.

4.8 ORDER_ITEM

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order identifier
Product_ID	Purchased product
Quantity	Quantity ordered
Unit_Price	Price per item

4.8.1 Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Unit_Price

4.8.2 Redundancy Analysis

- Order item stores only purchased products.
- Product information should not be duplicated here except the unit price at purchase time.

4.8.3 Data Anomalies

Insertion Anomaly

- Order items cannot exist without an order and product.

Update Anomaly

- Incorrect product references may create inconsistencies.

Deletion Anomaly

- Deleting order items removes purchase history.

4.9 PAYMENT

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Related order
Payment_Method	UPI, Card, Wallet
Payment_Date	Date of payment
Amount	Paid amount
Payment_Status	Success, Failed, Pending

4.9.1 Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Date, Amount, Payment_Status

4.9.2 Redundancy Analysis

- Payment details should be stored once for each order.
- Avoid repeating payment information in the Order table.

4.9.3 Data Anomalies

Insertion Anomaly

- Payment cannot be created before an order exists.

Update Anomaly

- Updating payment status in multiple places causes inconsistency.

Deletion Anomaly

- Deleting payment records removes transaction history.

4.10 SHIPPING

Attribute	Description
Shipping_ID	Unique shipping ID
Order_ID	Related order
Courier_Name	Delivery company
Tracking_Number	Tracking number
Shipping_Date	Shipping date
Delivery_Date	Delivery date
Shipping_Status	Shipping status

4.10.1 Functional Dependency

Shipping_ID → **Order_ID**, **Courier_Name**, **Tracking_Number**, **Shipping_Date**, **Delivery_Date**, **Shipping_Status**

4.10.2 Redundancy Analysis

- Shipping details should be maintained in one table.
- Orders should reference shipping records through **Order_ID**.

4.10.3 Data Anomalies

Insertion Anomaly

- Shipping record cannot exist without an order.

Update Anomaly

- Tracking information stored in multiple places causes inconsistency.

Deletion Anomaly

- Removing shipping records removes delivery history.

4.11 REVIEW

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer who reviewed
Product_ID	Reviewed product
Rating	Rating (1–5)
Comment	Review text
Review_Date	Review date

4.11.1 Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Comment, Review_Date

4.11.2 Redundancy Analysis

- Reviews should be stored only once.
- Customer and product details should be referenced using foreign keys.

4.11.3 Data Anomalies

Insertion Anomaly

- Review cannot exist without a customer and product.

Update Anomaly

- Updating product or customer details inside reviews creates redundancy.

Deletion Anomaly

- Deleting reviews removes customer feedback history.

4.12 WISHLIST

Attribute	Description
Wishlist_ID	Unique wishlist ID
Customer_ID	Customer ID
Product_ID	Saved product
Added_Date	Date added

4.12.1 Functional Dependency

Wishlist_ID → Customer_ID, Product_ID, Added_Date

4.12.2 Redundancy Analysis

- Wishlist stores only references to products.
- Product information should not be duplicated.

4.12.3 Data Anomalies

Insertion Anomaly

- Wishlist cannot exist without a customer and product.

Update Anomaly

- Duplicate wishlist entries create unnecessary redundancy.

Deletion Anomaly

- Removing wishlist records removes saved product information.

4.13 CART

Attribute	Description
Cart_ID	Unique cart ID
Customer_ID	Customer ID
Product_ID	Product ID
Quantity	Quantity selected
Added_Date	Date added

4.13.1 Functional Dependency

Cart_ID → Customer_ID, Product_ID, Quantity, Added_Date

4.13.2 Redundancy Analysis

- Cart stores only selected products.
- Product details remain in the Product table.

4.13.3 Data Anomalies

Insertion Anomaly

- Cart item cannot be added without a customer and product.

Update Anomaly

- Duplicate cart records create inconsistent quantities.

Deletion Anomaly

- Removing cart records deletes the customer's selected items.

4.14 ADMIN

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address
Password	Login password
Role	Administrator role

4.14.1 Functional Dependency

Admin_ID → Admin_Name, Email, Password, Role

4.14.2 Redundancy Analysis

- Administrator information should be maintained in one table.
- No administrator details should be duplicated in other tables.

4.14.3 Data Anomalies

Insertion Anomaly

- Administrator account cannot be created if mandatory fields are missing.

Update Anomaly

- Updating administrator information in multiple locations causes inconsistencies.

Deletion Anomaly

- Deleting an administrator record may remove administrative history if audit logs are not maintained.

4.15 CONCLUSION

This chapter identified the functional dependencies for each entity in the Etsy Marketplace Database System and examined possible redundancy and data anomalies. Proper use of primary keys and foreign keys ensures that each attribute depends on the correct key, reducing duplicate data and maintaining consistency. The analysis of insertion, update, and deletion anomalies highlights the importance of a well-designed relational database. These findings provide the foundation for the next stage of database normalization, where the schema will be refined to eliminate redundancy and improve overall database efficiency.